Rational Number Project

Level 2 / Lesson 5 / Overview

Students extend their fraction concepts by reconstructing the unit given the fractional part.

Materials

- Cuisenaire rods for students and teacher
- Student Pages A-C

Teaching Actions

1. Lead a discussion around the concept of flexibilty of unit. Possible questions include:

   (a) What cuisenaire rod can be used as the unit to model 1/3? Is that the only rod?

   (b) What criteria do you use to determine if a rod can be a unit for 1/3? Why?

   (c) Show 1 red rod and ask: Can this rod represent the fraction 1/2 and 1/4? How is this possible?

2. Explain that so far they have used cuisenaire rods to model fractions. A unit was determined, partitioned into equal parts, and then a certain number of parts were looked at.

3. Review with these examples:

   (a) Show 2/3 with the blue rod as the unit. Why does this show 2/3?

   (b) Purple is what fraction of the brown? Why?

   (c) What rod is 1/3 of the blue rod? Why?

Comments

1. In Level 1, lesson 16, students learned to reconstruct the unit given the fraction using fraction circles. This lesson builds on the same concept but now students use rods as the model.

2. Initial discussion revolves around <u>two</u> main ideas. First, different units can be used to model the same fraction. Second, the fractional value of a rod is determined by the unit to which it is compared.

   [r] = 1/2 if the unit is [purple]

   [r] = 1/4 if the unit is [brown]

Teaching Actions

4. Explain that you are going to change the question once more.

   LeAnna ate 1/3 of a candy bar. The amount she ate was the same size as 1 green. Find the size of the whole candy bar.

5. Step back and have students in pairs solve the problem using their rods. Have students explain their solution process with the rods. If needed, help the children verbalize by asking these questions:

   (1) Will the whole candy bar be bigger or smaller than 1 green?

   (2) How many 1/3's make up the unit?

   (3) The green is 1/3 of what rod?

6. Display the red rod and say this is 1/4. With your partner, find the unit and talk aloud how this is done.

7. Share responses in large group. Repeat for:

   purple is $\frac{1}{2}$

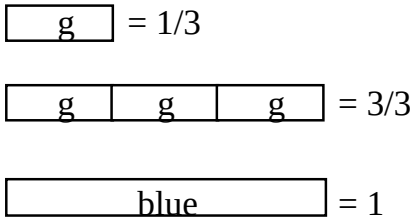
   red is $\frac{1}{5}$

   white is $\frac{1}{5}$

   purple is $\frac{1}{3}$

Comments

3. If the fraction is 1/3, then three, 1-thirds equal the unit.



4. Continue to encourage students' verbalization of their actions on the rods by having students do problems in pairs.

Teaching Actions

8. Now present this problem:

   LeAnna ate 2/3 of the candy bar. The amount she ate was the same size as 1 purple. Find the size of the whole candy bar.

9. Again, step back and see how students solve the problem using their rods. Have students explain their solution process with the rods. Leading questions if needed are:

   (1) If the purple is 2/3 of some unit, would the unit be longer, the same size, or shorter than the purple?

   (2) If the purple is 2/3, would 1/3 be longer, shorter, or the same size?

   (3) What would 1/3 of the unit be?

   (4) If red is 1/3, what is the unit?

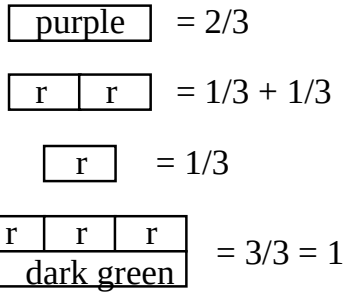
   (5) Can you verify that the purple is 2/3 of the dark green?

10. Student Pages A-C provide practice with finding the unit given the fractional part.

Comments

5. To solve this type of problem students need to find the **unit fraction** first, and then reconstruct the unit as done previously.

   Example:

6. Problem 4 on Student Page A is challenging. Let the students struggle a bit with these examples before giving them assistance.

   To solve the problem the student has to determine the unit, and then consider each other rod in relationship to the unit.

   Example:

   If one purple equals 1/2 then two purples or 1 brown equals the unit. Therefore:

   1 white is 1/8 of the brown
   1 red is 2/8 or 1/4 of the brown.
   1 yellow is 5/8 of the brown
   1 brown = 1 or the whole

Find the Unit Using the Rods

1. Red is 1/2 of some unit. Find the unit. Draw a picture of the unit.

2. Green is 1/4 of some unit. Find the unit. Draw a picture of the unit.

3. White is 1/8 of some unit. Find the unit. Draw a picture of the unit.

4. If the purple rod is 1/2 of some unit, what value do each of these rods have?

   1 white = _____ 1 red = _____

   1 yellow = _____ 1 brown = _____

Directions:
Solve these problems using your Cuisenaire Rods.

1. This is $\frac{2}{3}$ of the unit.

So this is _____ of _____ equal-size parts of the unit.

Draw the unit.

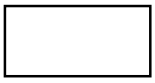
Tell which rod or rods make the unit. _____

2. This is $\frac{1}{2}$ of the unit.

So this is _____ of _____ equal-size parts of the unit.

Draw the unit.

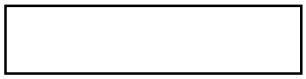
Tell which rod or rods make the unit. _____

3.  This is $\frac{2}{5}$ of the unit.

So this is _____ of _____ equal-size parts of the unit.

Draw the unit.

Tell which rod or rods make the unit. _____

4.  This is $\frac{4}{7}$ of the unit.

So this is _____ of _____ equal-size parts of the unit.

Draw the unit.

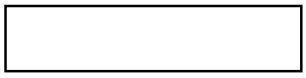
Tell which rod or rods make the unit. _____

5. This is $\frac{3}{8}$ of the unit.

So this is _____ of _____ equal-size parts of the unit.

Draw the unit.

Tell which rod or rods make the unit. _____

6.  This is $\frac{2}{4}$ of the unit.

So this is _____ of _____ equal-size parts of the unit.

Draw the unit.

Tell which rod or rods make the unit. _____